

The **energese** Package User Guide

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1 Introduction

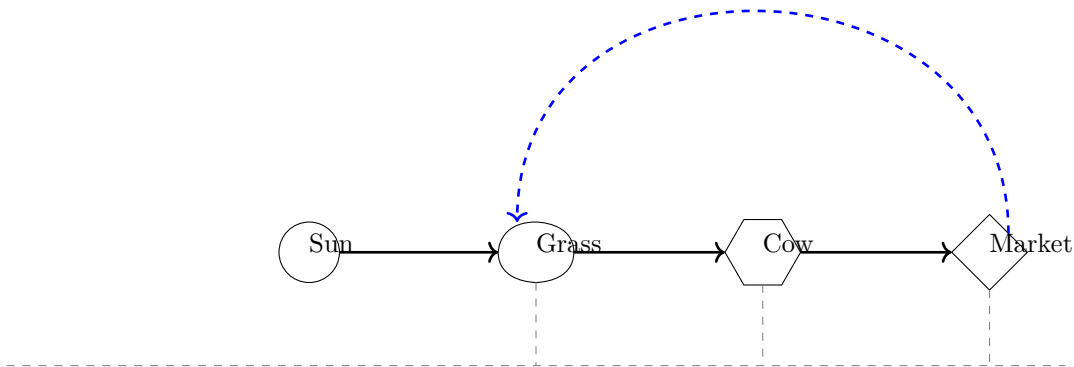
The **energese** package automates the rendering of H.T. Odum's Energy Systems Language (ESL) diagrams using a data-driven approach via LuaLaTeX and TikZ.

2 Quick Start

To use the package, you need to provide a JSON file describing your system.

2.1 Example System

Here is a simple ESL diagram rendered from a JSON file.



2.2 JSON Structure

The JSON file should contain **nodes** and **edges**. Each node must have a **Tr** (Transformity) value, which determines its horizontal position.

```
{
  "nodes": [
    {"id": "sun", "type": "source", "Tr": 1, "label": "Sun"},
    {"id": "grass", "type": "producer", "Tr": 100, "label": "Grass"}
  ],
  "edges": [
    {"from": "sun", "to": "grass", "type": "energy"}
  ]
}
```

3 Node Types

The package supports several node types:

- **source:** Circle
- **producer:** Lung shape
- **consumer:** Hexagon
- **storage:** Tank
- **interaction:** Pointed block
- **transaction:** Diamond

4 Edge Types

- **energy:** Solid line with arrow
- **money_feedback:** Dashed blue overhead arc
- **information:** Thin solid line
- **material:** Solid line